

SIXTH SEMESTER DIPLOMA EXAMINATION IN ENGINEERING/ TECHNOLOGY

ANALYTICAL INSTRUMENTS (4088)

Model question paper

(Maximum marks: 100)

PART-A

I Answer the following questions in one or two sentences. Each question carries 2 marks.

1. What do you mean by Emission spectrum?
2. Name the detectors used in Infrared Spectrophotometer.
3. List the sources of errors in Glass electrode.
4. State the principle of Paramagnetic analyser.
5. List the common water pollutants.

PART- B

II Answer any *five* of the following. Each question carries 6 marks.

1. State and explain the fundamental laws of Photometry.
2. Explain the working principle of Flamephotometer.
3. Explain the importance of Buffer solution.
4. Describe the working of digital pH meter.
5. Explain the construction and operation of Thermal conductivity analyser.
6. Explain the method of effluent treatment.
7. Explain the Electrostatic precipitator in pollution control.

PART- C

(Answer one full question from each unit. Each question carries 15 marks)

UNIT-I

III With neat diagrams, Explain the construction and principle of

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|-----------------------------------|---|
| (a) Single beam filter photometer | 8 |
| (b) Double beam filter photometer | 7 |

OR

IV (a) State and explain the principle of Raman Effect. 5

(b) Describe the construction and working of N.M.R spectrometer 10

UNIT-II

- V (a) State and explain the importance of Chromatographic process. 5
- (b) With a neat block schematic, explain the construction and working of Gas Chromatography. 10

OR

VI Write short notes on

- (a) Hydrogen electrode 8
- (b) Glass electrode 7

UNIT-III

VII Explain the construction and operation of

- (a) Infrared gas analyser 8
- (b) Electrical conductivity analyser 7

OR

VIII Describe the construction and operation of

- (a) Paramagnetic oxygen analyser 8
- (b) Zirconia oxygen analyser 7

UNIT-IV

- IX (a) Specify the need for noise measurement in industry. 8
- (b) Explain the industrial noise control methods. 7

OR

- X (a) Describe the method of flue gas analysis. 8
- (b) Explain the method of monitoring of carbon monoxide. 7